Altair AI Studio – Second Screenshot Notes

Line-by-line transcription and explanation of the four uploaded screenshots

1. Combined Decision Tree and Random Forest Workflow

• Application: Altair AI Studio Educational 2026.1.1.

• Process input: Read CSV.

• Read CSV output is connected to the Multiply operator input (inp).

• Multiply has one input (inp) and multiple outputs (out).

• One Multiply output is connected to the Decision Tree training input (tra).

• Another Multiply output is connected to the Random Forest training input (tra).

• Decision Tree outputs: model (mod), example set (exa), and weights (wei).

• Random Forest outputs: model (mod), example set (exa), and weights (wei).

• The model and example-set outputs are routed to result ports on the right.

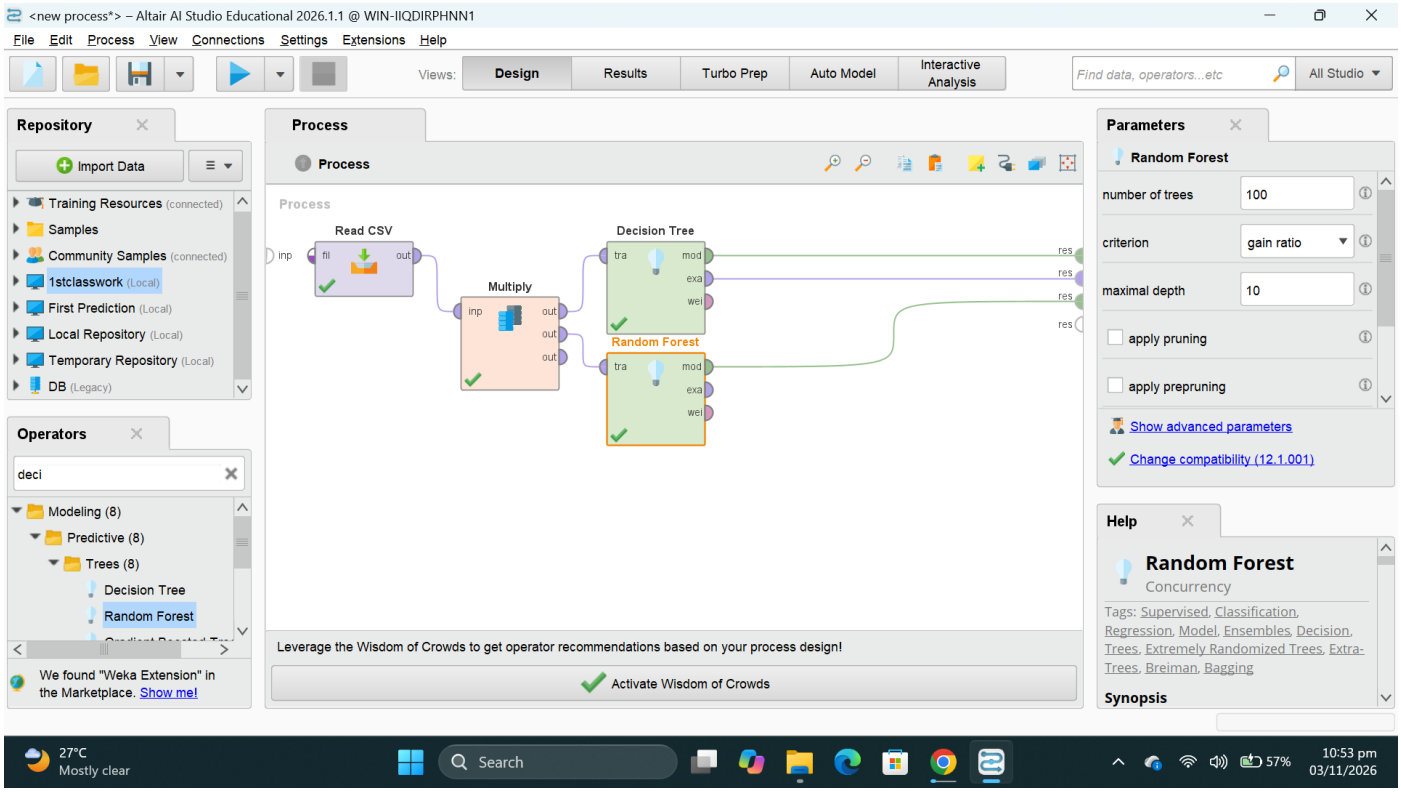
• The Random Forest operator is selected in the process.

• Random Forest parameters: number of trees = 100; criterion = gain ratio; maximal depth = 10.

• Apply pruning is unchecked; apply prepruning is unchecked.

• Operator search field contains: deci.

• Operator category shown: Modeling → Predictive → Trees.



Explanation: The CSV data is passed into Multiply so copies of the same dataset can be supplied to two learning algorithms. Decision Tree and Random Forest can then be compared.

2. Decision Tree – Shape Classification

• Result tab: Tree (Decision Tree).

• Display type: Tree. Node Labels and Edge Labels are checked.

• Root attribute: Color.

• Color = green → next attribute: Outline.

• Color = red → next attribute: Dot.

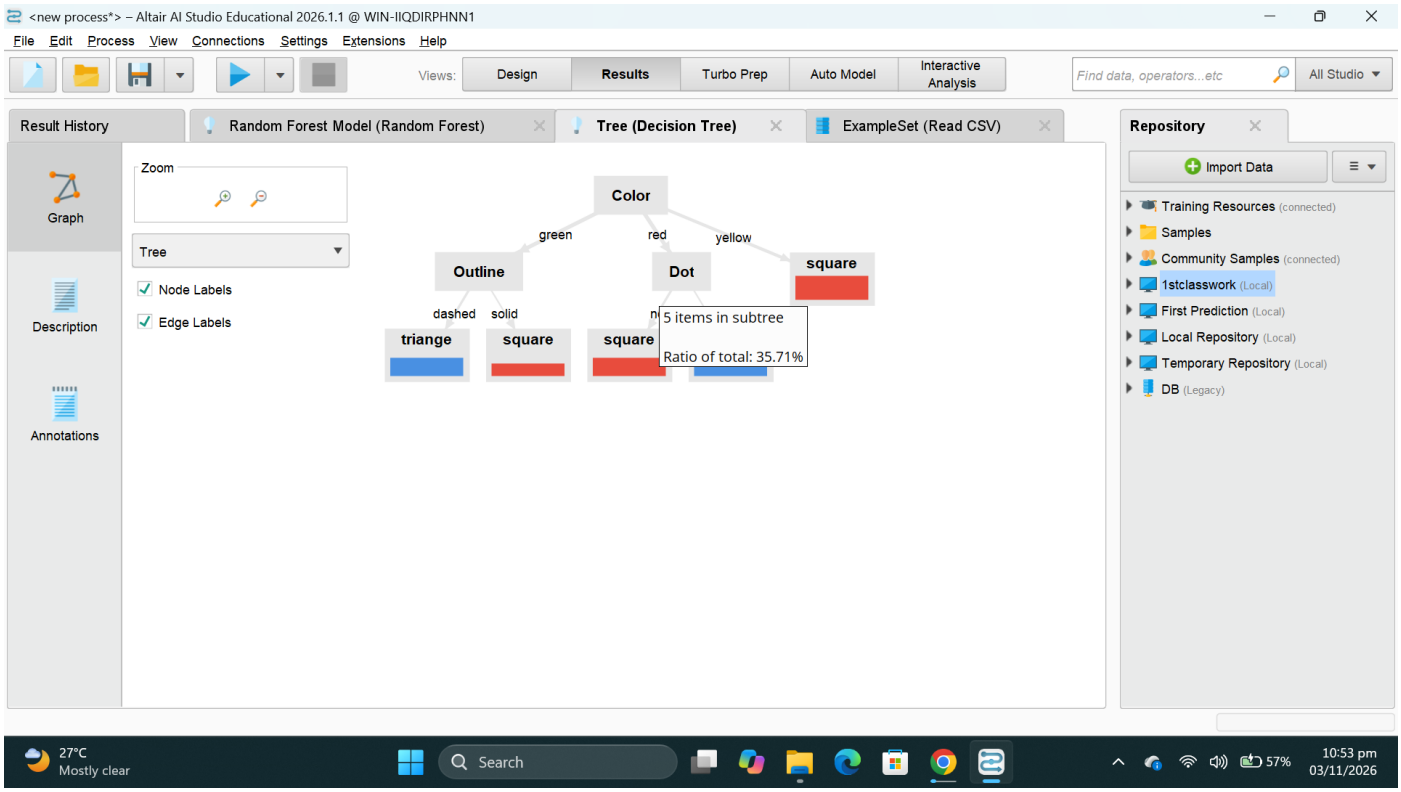
• Color = yellow → predicted class: square.

• Under Outline: dashed → triangle; solid → square.

• Under Dot: no → square; yes → triangle.

• Visible tooltip: 5 items in subtree; Ratio of total: 35.71%.

• The colored bars under leaf labels visually represent the predicted class.



Explanation: This is a classification tree. It predicts a shape (triangle or square) by checking Color first, then Outline or Dot where needed.

3. Random Forest Model – Individual Tree

• Result tab: Random Forest Model (Random Forest).

• The left Result History panel contains multiple entries named Tree.

• One Tree entry is selected.

• Display type: Tree. Node Labels and Edge Labels are checked.

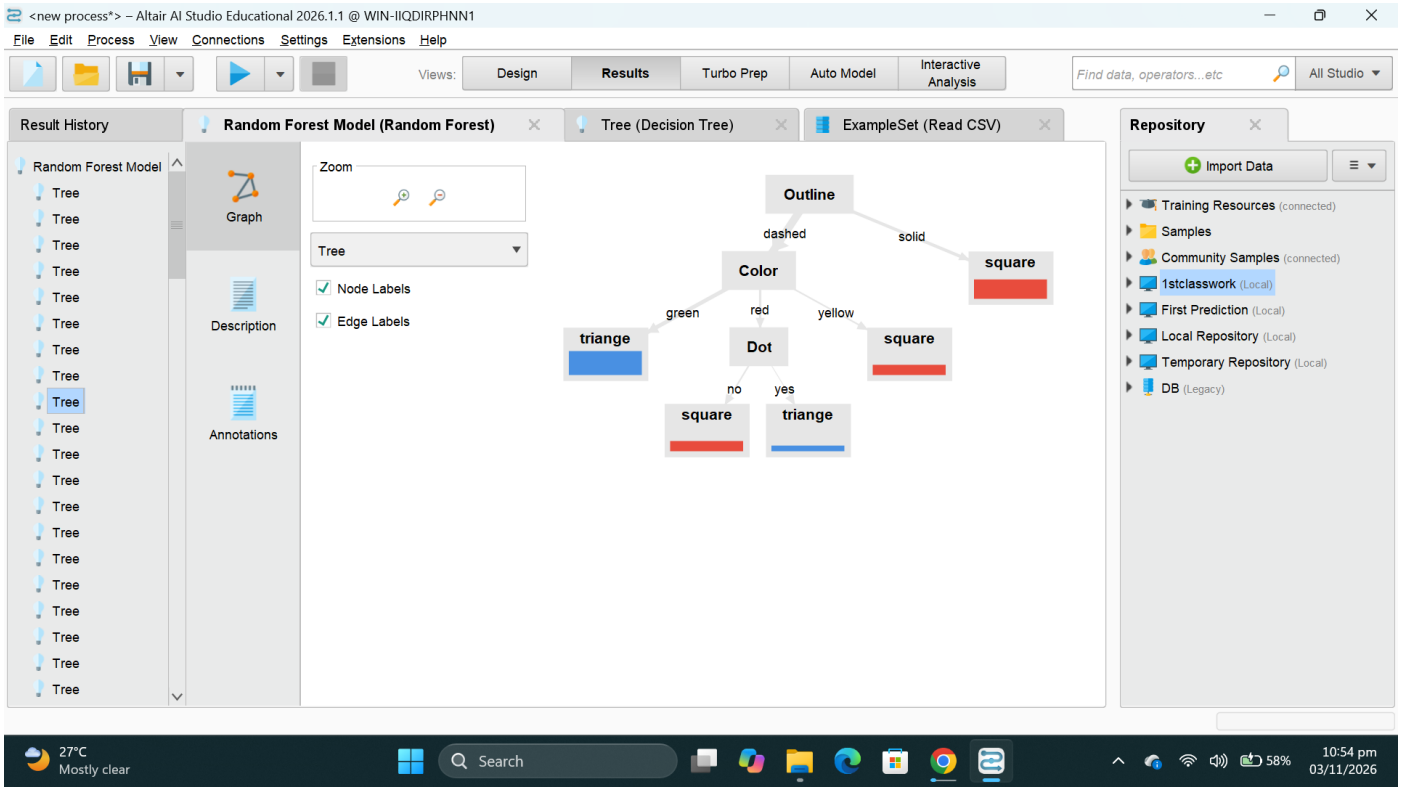
• Root attribute: Outline.

• Outline = dashed → next attribute: Color.

• Outline = solid → predicted class: square.

• Under Color: green → triangle; red → next attribute: Dot; yellow → square.

• Under Dot: no → square; yes → triangle.



Explanation: A Random Forest contains multiple decision trees. This screenshot shows one tree from the forest; different trees can use different attributes and rules.

4. Decision Tree Result and Dataset Text

• The Altair result view displays the Decision Tree for shape classification.

• Root attribute: Color.

• Color = green → Outline; dashed → triangle; solid → square.

• Color = red → Dot; no → square; yes → triangle.

• Color = yellow → square.

• A text editor window is open with the file name: dataset-shapes.csv.

• Visible CSV rows: yellow,dashed,yes,square

• Visible CSV row: yellow,solid,no,square

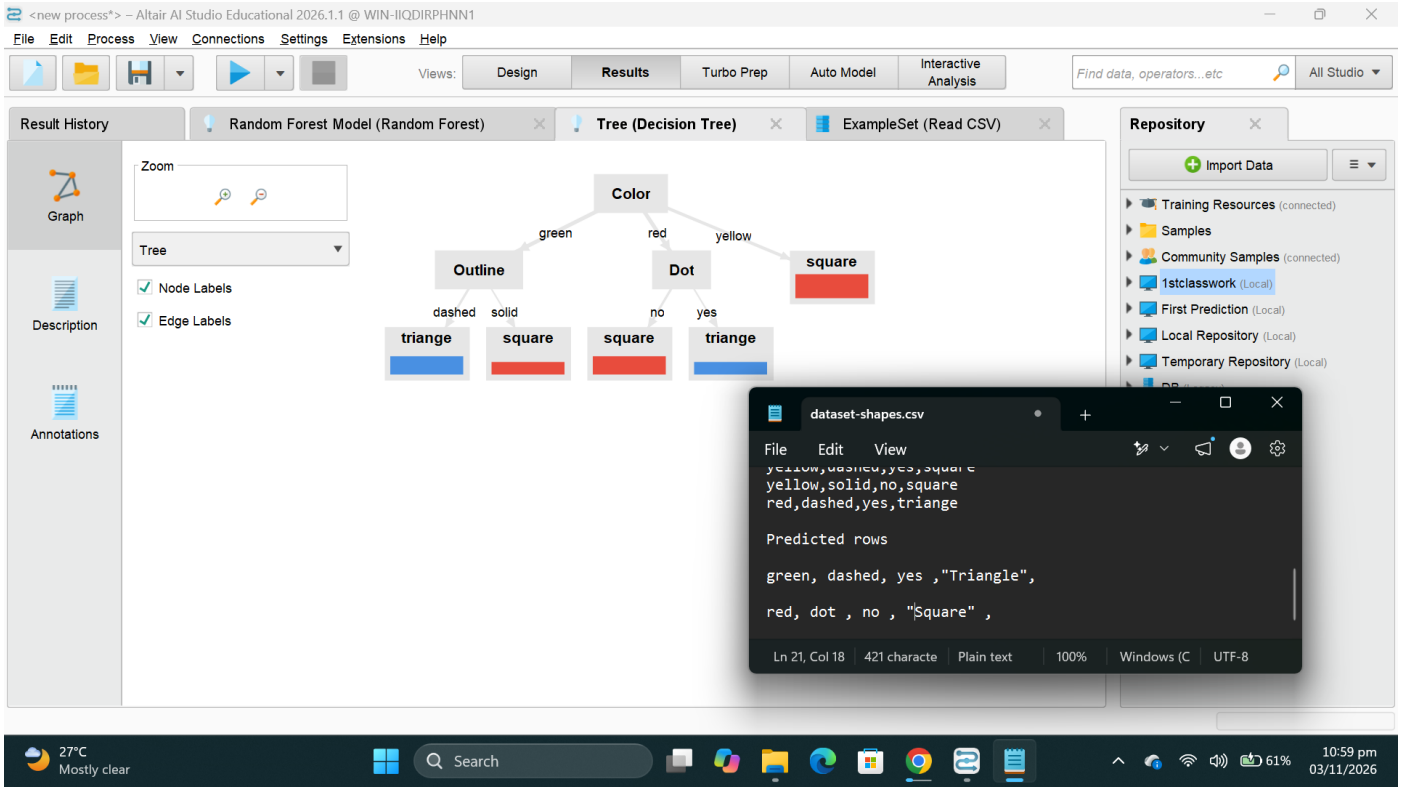
• Visible CSV row: red,dashed,yes,triangle

• A section heading in the file reads: Predicted rows.

• First visible predicted row: green, dashed, yes, "Triangle",

• Second visible predicted row: red, dot, no, "Square",

• Text editor status: Ln 21, Col 18; 421 characters; Plain text; 100%; Windows (CRLF); UTF-8.



Explanation: The CSV contains features such as Color, Outline, and Dot, along with a shape label. The tree learns rules from examples and predicts a shape for new rows. Only the visible part of the CSV has been transcribed.

Overall Summary

These screenshots demonstrate classification in Altair AI Studio. A Decision Tree predicts shapes from features such as color, outline, and dot. A Random Forest combines many decision trees. The Multiply operator sends copies of the same input data to more than one model.